

Discussion 2 Solutions

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Problem 1:

- (i) Fill in the table

p	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg p \vee \neg q$	$\neg q \wedge \neg p$	$p \Rightarrow q$	$\neg p \Rightarrow \neg q$	$\neg q \Rightarrow p$
T	T	F	F	T	F	F	T	T	T
T	F	F	T	F	T	F	F	T	T
F	T	T	F	F	T	F	T	F	T
F	F	T	T	F	T	T	T	T	F

- (ii) Identify if the following statements are true using the truth table above (extend the truth table if necessary)

- (a) $(p \wedge q) \Leftrightarrow \neg(\neg p \vee \neg q)$ **True**
(b) $(p \Rightarrow q) \Leftrightarrow (\neg p \Rightarrow \neg q)$ **False**
(c) $\neg(\neg q \Rightarrow p) \Leftrightarrow (\neg q \wedge \neg p)$ **True**

Problem 2: Determine the logical negation of each claim.

- (i) There exists a country where every city has a university.

Solution: In every country, there exists a city without a university.

- (ii) For all positive real numbers x , $-x < 0$.

Solution: There exists a positive real number x such that $-x \geq 0$

- (iii) $\exists x \in \mathbb{R} : \forall y, z \in \mathbb{Z}, \neg P(x, y, z)$

Solution:

- $\neg(\exists x \in \mathbb{R} : \forall y, z \in \mathbb{Z}, \neg P(x, y, z))$
- $\forall x \in \mathbb{R}, \neg(\forall y, z \in \mathbb{Z}, \neg P(x, y, z))$
- $\forall x \in \mathbb{R}, \exists y, z \in \mathbb{Z} : \neg(\neg P(x, y, z))$
- $\forall x \in \mathbb{R}, \exists y, z \in \mathbb{Z} : P(x, y, z)$

(iv) $\forall x, y \in \mathbb{R} : (x < y \Rightarrow \exists q \in \mathbb{Q} : x < q < y)$

Solution:

- $\neg(\forall x, y \in \mathbb{R} : (x < y \Rightarrow \exists q \in \mathbb{Q} : x < q < y))$
- $\exists x, y \in \mathbb{R} : \neg(x < y \Rightarrow \exists q \in \mathbb{Q} : x < q < y)$
- $\exists x, y \in \mathbb{R} : (x < y) \wedge \neg(\exists q \in \mathbb{Q} : x < q < y)$
- $\exists x, y \in \mathbb{R} : (x < y) \wedge (\forall q \in \mathbb{Q}, q \leq x \vee q \geq y)$

Problem 3: Write the negation, converse, inverse, and contrapositive of each of the following statements.

(i) Every polygon that is a square has four sides.

- (a) Negation: There exists a polygon that is a square and does not have four sides.
- (b) Converse: If a polygon has four sides, then it is a square.
- (c) Inverse: If a polygon is not a square, then it does not have four sides.
- (d) Contrapositive: If a polygon does not have four sides, then it is not a square.

(ii) $\forall x \in \mathbb{R}, (x > 2 \Rightarrow x^2 > 4)$

- (a) Negation: $\exists x \in \mathbb{R} : (x > 2 \wedge x^2 \leq 4)$
- (b) Converse: $\forall x \in \mathbb{R}, (x^2 > 4 \Rightarrow x > 2)$
- (c) Inverse: $\forall x \in \mathbb{R}, (x \leq 2 \Rightarrow x^2 \leq 4)$
- (d) Contrapositive: $\forall x \in \mathbb{R}, (x^2 \leq 4 \Rightarrow x \leq 2)$

Problem 4: For each of the following, determine whether P and Q are logically equivalent (i.e., whether $P \Leftrightarrow Q$ is true).

(a) Given

$$P : 2x - 3 = 7, \quad Q : x > 4$$

Solution:

- $P \Rightarrow Q : x = 5 \Rightarrow x > 4$
If P is true, then $x = 5$. Since $5 > 4$, Q is also true. Hence $P \Rightarrow Q$ is **True**.
- $Q \Rightarrow P : x > 4 \Rightarrow x = 5$
If Q is true, then $x > 4$. However, this does not guarantee that $x = 5$. For example, when $x = 6$, Q is true but P is false. Hence $Q \Rightarrow P$ is **False**.
- Therefore, $P \Leftrightarrow Q$ is **False**.

(b) Let X and Y be arbitrary sets. Given

$$P : X \subseteq Y, \quad Q : X \cup Y = Y$$

Solution:

- If $X \subseteq Y$, then every element of X is already in Y . Thus adding X to Y does not change Y , so $X \cup Y = Y$. Therefore, $P \Rightarrow Q$ is always **True**.
- Conversely, if $X \cup Y = Y$, then every element of X is already in Y . Hence, $X \subseteq Y$, so $Q \Rightarrow P$ is also always **True**.
- Therefore, $P \Leftrightarrow Q$ is **True**.

Problem 5: Define perfect squares using Biconditional statement.

Solution: $n \in \mathbb{Z}$ is a perfect square $\Leftrightarrow \exists k \in \mathbb{Z} : n = k^2$.